Inner Tracker: What needs calibrating?

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summarizing the work of many

CMS Tracker DPG Workshop — 14–16 October 2026

Summary

Day 1:

- What needs to be calibrated, and why
- Each effect moves several physics variables at once

Evolution of the detector:

- Radiation degrades the chip and the sensor gradually, in ways we can (partially) predict
- Handles to mitigate them: threshold, timing, HV, front-end biases, offline CPE

Proposal:

- A calibration programme driven by monitored observables
- To discuss: tolerances from simulation, and observables that tell us when to recalibrate

Current status of the simulation

Joint work of Suchandra, Stian and Michelangelo

- **Before:** ideal front-end in CMSSW → rise time 0 and perfect charge-to-ToT conversion, no sampling logic or full timing effects, partial (and CPU-slow) time-walk addition
- **Now:** realistic CROC waveform ("waveform 2.0"): full waveform model built using dedicated signal scans on real chips. Also added CROC sampling logic for a fully realistic event timing model. Currently in a CMSSW branch, being merged
- One parameter for the delay between LHC clock and chip clock → the one we tune in operation for best timing alignment
- Noise also generated in modules without a hit

⇒ we can simulate the calibration procedures themselves (e.g. fine delay scan with 1 colliding bunch)

- **First results (Pari, Workshop on IT BES requirements):** tracking with waveform 1.0 (low time-walk), re-run with 2.0 ongoing

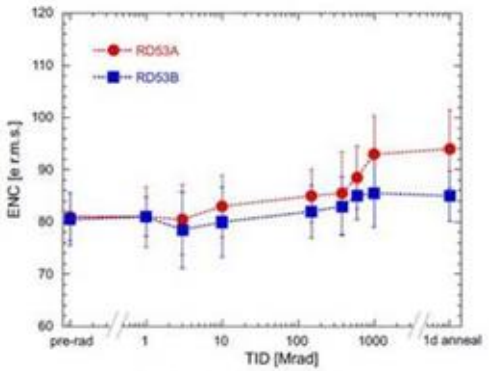
Details: S. Mæhre, Studies of IT RD53 timing effects in Phase 2

Physics variables of interest

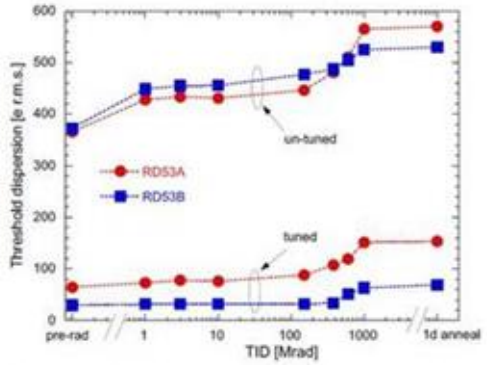
	Noise rate	Hit efficiency	Out-of-time fraction	Cluster resolution	Collected charge (ToT)
Definition	Rate of hits without a particle, per pixel per BX	Fraction of tracks that leave a hit	Hits assigned to the wrong BX	Position error of a cluster	Charge per hit, in 25 ns counts
What moves it	Threshold, ENC (Equivalent Noise Charge)	Threshold, timing, charge collection, masked/dead pixels	Time-walk, clock phase, Krummenacher	Cluster size, charge sharing, threshold, CPE	Charge collection, Krummenacher, threshold
How we see it	Noise scan, occupancy	On-track efficiency (DQM)	Min-bias cluster count, timing scan	Track residuals	ToT spectrum, cluster charge

How the chip degrades with radiation

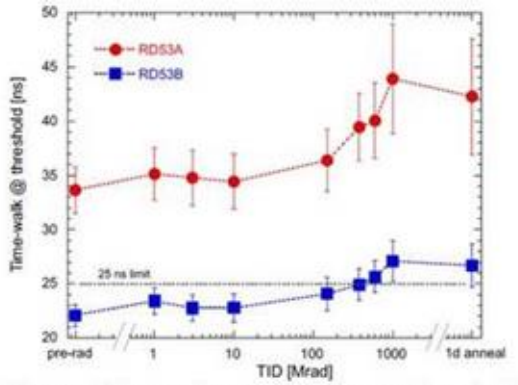
	ENC	Un-tuned threshold dispersion	Tuned threshold dispersion	Time-walk at threshold	Bias and references (IREF, VREF, VDDA/D)	Digital logic delay
Pre-rad	~85 e	~370 e	~40 e	~22 ns	nominal	1×
1 Grad	~+6% (< 10 e)	~530 e	~70 e	~27 ns	+10–15% (X-ray; not seen with γ)	~2×
Consequence	Small: sensor leakage dominates	Re-tune at 0.5, 1, 5, 10, 50, 100 Mrad, then every 100 Mrad	Low-threshold pixels make the noise: keep $\rho = \sqrt{(ENC^2 + \sigma^2)} <$ 126 e (RD53 specs)	Optimal clock phase moves	VDD re-trim, ADC correction	Design margin, no action



Equivalent Noise Charge



Un-tuned and tuned threshold dispersion



Time-walk for an input signal close to threshold

[Plots and table: Loddo, IT BES workshop, slides 4 and 6]

How the sensor degrades with radiation

	Charge per MIP (MPV)	Leakage current	Bias for full efficiency (>99% planar, ~97% 3D)	Threshold for 10^{-6} noise occupancy ≈ 4.75σ
Unirradiated (150 µm thick)	~11 ke	< 0.75 µA/cm² (spec, 20 °C)	≤ 10 V planar, ≤ 5 V 3D	~450 e (ENC 85, σ 40)
3D (L1), irradiated 1–1.6×10¹⁶ n_eq/cm²	~5 ke at 1×10¹⁶ (130 V)	not reported	~30–50 V for 97%	~670 e (ENC 122 at 10 nA + 1 Grad, σ 70)
Planar (L2+, TFPX), irradiated 5×10¹⁵–2×10¹⁶ n_eq/cm²	not reported	≤ 45 µA/cm² at 5×10¹⁵, 600 V, −25 °C (spec): ≲ 1 nA/pixel	350 V at 5×10¹⁵; < 800 V up to 2×10¹⁶	~560 e (ENC ~94 at 1 nA + 1 Grad, σ 70)
Consequence	Fewer ToT counts per hit: re-check gain	ENC 85 → 115 e: shot noise of 10 nA/pixel leakage	HV scans track the plateau (HV handle slide)	Gaussian estimate, below the 1000 e CMS target: margin for non-Gaussian tails

Table: 3D, arXiv:2502.05521 (CROC); planar, arXiv:2307.01580 (RD53A); ENC: Loddo, IT BES workshop, slide 5

Each effect moves several variables

	Noise rate	Efficiency	Out-of-time fraction	Hit resolution	Charge (ToT)
Noise (ENC) ↑	●	○			○
Threshold dispersion ↑	●	●	○	○	
Time-walk ↑		●	●	○	
Charge collection ↓		●	●	●	●
Leakage current ↑	●	○			○
Bias / reference drift	○	○	○		●

● direct effect ○ indirect effect. Draft for discussion: to check against simulation and test-beam data.

Before the handles: charge and ToT

- ToT counts the 25 ns clocks the signal stays above threshold; the Krummenacher current sets the return to baseline
- **One ToT count ≈ 3 ke in L1, ≈ 1.5 ke elsewhere**
- Higher Krummenacher current: faster discharge, more charge per count
- The ToT offset moves with the threshold: **every threshold retune needs a new gain scan**
- Noise lands in ToT 0; the ToT spectrum tracks charge collection

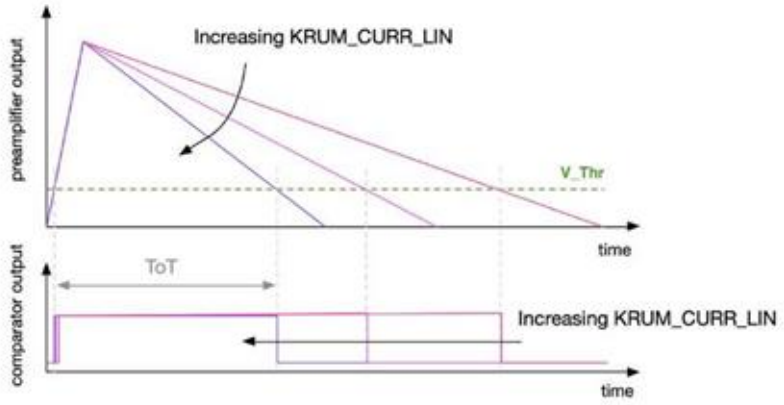
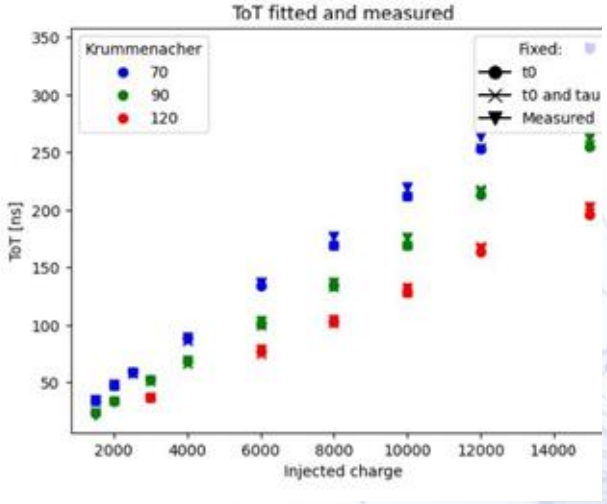


Figure 2.7: Effects of Krummenacher current on signal waveform and ToT.

Damenti, MSc thesis, 2023, Fig. 2.7



Mæhre and Pari, IT BES workshop, slide 37

Handle: threshold (GDAC + per-pixel TDAC)

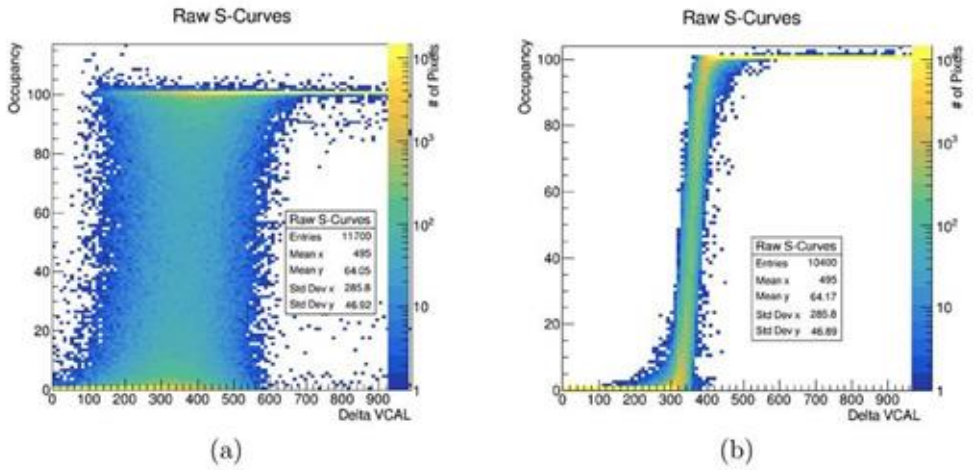
Threshold ↑	Noise rate	Hit efficiency	Out-of-time fraction	Single-hit resolution	Collected charge (ToT)
Effect	+++	−−	+	−	↻
Why	Gaussian tail cut	Small hits lost	Less overdrive: time-walk	Edge pixels lost, clusters break	ToT offset moves: new gain scan

+ better · − worse · count = strength · – no effect · ? to check · ↻ redo calibration. Magnitudes are a draft.

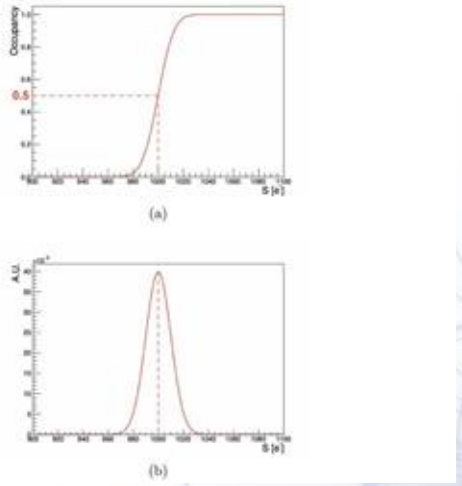
- Set by GDAC (global, per front-end region) and TDAC (per pixel, range set by LDAC), in electrons via VCAL injection
- Phase-1 policy: raise it when forced, never lower it opportunistically
- **Do we keep this policy for Phase-2?**
- Goal: the lowest threshold that keeps noise inside an agreed budget
- **A MIP leaves ~11 ke in 150 µm, shared among the pixels of its cluster: edge pixels collect only a few ke**
- **After 1×10^{16} the 3D signal drops to ~5 ke, so many pixels sit close to a 1 ke threshold**
- **Raising the threshold drops them: smaller clusters, worse resolution, lost hits. 3D at 1.6×10^{16}: 2× threshold costs up to 16% hit efficiency**

How we tune today: Ph2_ACF

- ThresholdAdjust: GDAC to the target, in steps (2000 → 1200 → 1000 e)
- ThresholdEqualization: per-pixel TDAC for the narrowest distribution; LDAC chosen for minimum σ_thr. Typical σ_thr: ~350–400 e untuned → 40–50 e tuned at 1 ke, 15 °C (Grippo, IT BES workshop, slide 15)
- InjectionDelay: set the injection timing so the threshold is measured in time
- Noise: mask pixels with noise occupancy above 10^{-4} (10^{-6} chip-wide)
- SCurve: measure threshold and noise of every pixel
- **Built for reception**: a narrow response, and mask whatever does not fit
- **TBPX L1 3D module at 1000 e: 1822 pixels flagged noisy after equalization**



Damenti, MSc thesis, 2023, Fig. 2.14



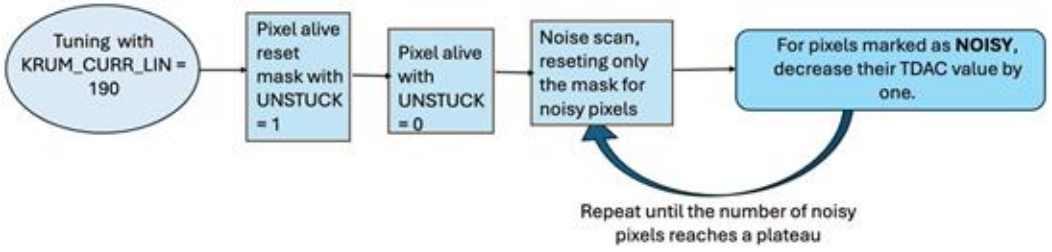
Damenti, MSc thesis, 2023, Fig. 2.12

Proposal: tune for noise

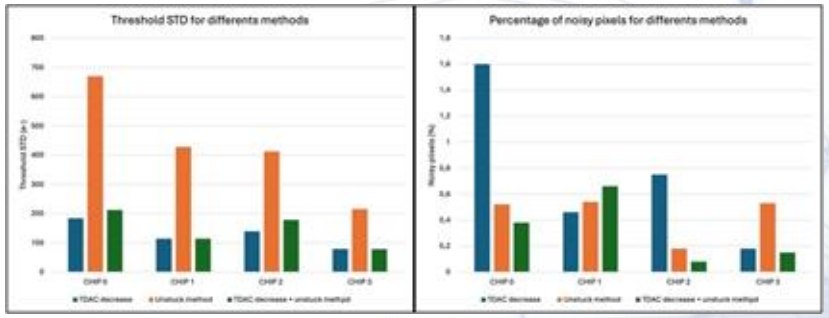
Target a noise level, not a charge: threshold down to the max acceptable noise → max efficiency

1. Set GDAC per front-end region, equalize once
2. Recover masked pixels: PixelAlive → noise scan →lower TDAC (higher threshold) for noisy pixels
3. Lower the threshold until noise reaches the **guard level per layer/ring**: 1/100 of the hit occupancy (1/1000 in L1).
L1: ~2×10^{-3} hits/pixel/BX → noise $\lesssim 2\times10^{-6}$. **Spread up to ~500 e is fine in L1** (below one ToT step, 1.5–3 ke)
4. Set the injection timing, run S-curves → in-time threshold of each pixel
5. **Re-run the gain (ToT) scan** after any threshold change

Similar approach: Gross and Bardelli, IT Modules meeting, 2 Jul 2026 → irradiated TBPX L1 module at 1000 e: **masked pixels 25.7k → 1.9k** (4 chips), σ_thr almost unchanged



Gross and Bardelli, IT Modules meeting, 2 Jul 2026, slide 13



Gross and Bardelli, IT Modules meeting, 2 Jul 2026, slide 16

Handle: per-chip delay (phase, latency)

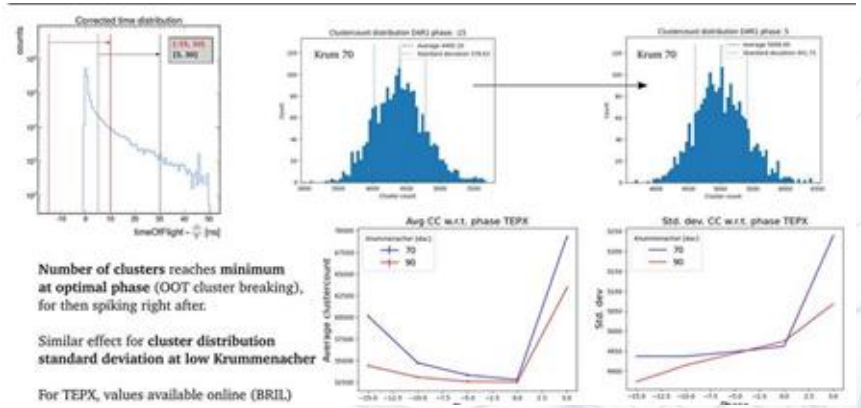
Delay → optimum	Noise rate	Hit efficiency	Out-of-time fraction	Single-hit resolution	Collected charge (ToT)
Effect	–	++	−−−	++	–
Why	No effect	Follows out-of-time fraction	Sets the sampling phase	Cluster-count minimum = best Y	No effect (sim.)

+ better · – worse · count = strength · – no effect · ? to check · ↻ redo calibration. Magnitudes are a draft.

- **Time-walk grows with dose**, so the optimal phase moves
- Cluster count in min bias has a minimum at the optimal phase and tracks ttbar Y-resolution

Proposal

- Commissioning: per-chip coarse + fine phase scan with collisions, pick the **cluster-count minimum**
- **Barrel monitor:** cluster count per module from DQM as the online observable (TEPX already has BRIL at 1 Hz)
- Re-scan when the monitor drifts and after every TID re-tune step
- **Open:** cable and fibre offsets in CROC latency or in DTC firmware?



Mœhre and Pari, IT BES workshop, slide 24

Handle: high voltage

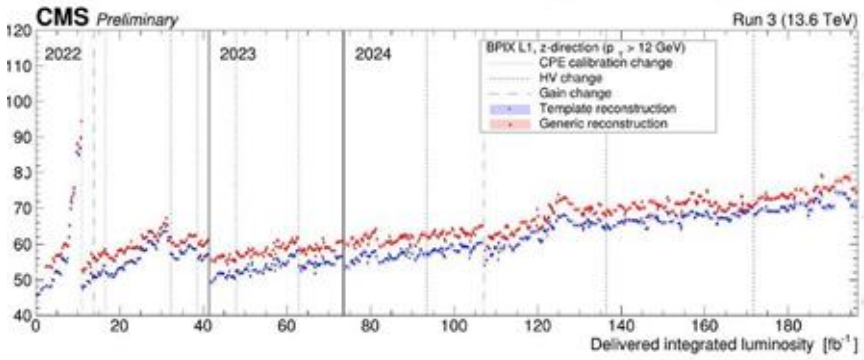
HV ↑	Noise rate	Hit efficiency	Out-of-time fraction	Single-hit resolution	Collected charge (ToT)
Effect		+++		++	+++
Why		Recovers irradiated sensors		Larger clusters	Recovers charge collection

+ better · – worse · count = strength · – no effect · ? to check · ↻ redo calibration. Magnitudes are a draft.

- Radiation traps charge in the sensor: more bias collects it again, so **the bias needed rises with fluence**
- HV scan: step the bias, measure cluster charge and efficiency, set HV on the plateau plus a margin
- Phase-1: full scan once a year (3–4 h of stable beams), mini scan every 20–30 fb^{-1} on a few power groups

Proposal

- Scan frequency per layer, scaled with fluence: **L1 most often**
- Use all clusters instead of on-track only: cheaper, but check that loopers do not bias the plateau
- **Open:** margin above the plateau vs leakage current



Roguljic (Tracker DPG), IT BES workshop, slide 4

Handle: Krummenacher current

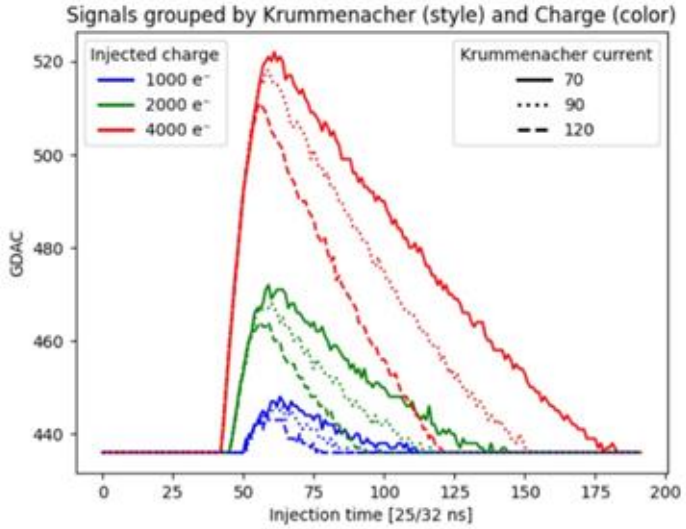
Krum ↑	Noise rate	Hit efficiency	Out-of-time fraction	Single-hit resolution	Collected charge (ToT)
Effect	–	+?	–	–?	– – – ↻
Why	No direct effect	Indirect: shorter ToT, less dead time	No direct effect	Indirect: coarser charge per ToT count	Shorter ToT; redo gain scan

+ better · – worse · count = strength · – no effect · ? to check · ↻ redo calibration. Magnitudes are a draft.

- Feedback current that discharges the preamplifier after a hit and sinks the sensor leakage: higher current, faster return to baseline, **shorter ToT**
- TBPX L1 3D modules run at 190 (default 80)
- Changing it moves the charge-to-ToT mapping: the gain scan must be redone
- **Today:** Ph2_ACF gainopt binary-searches DAC_KRUM_CURR_LIN so that 6 ke gives mean ToT 5.3 (slow) or 2 (fast discharge), after tuning to ~1 ke. Not in module QC

Proposal

- One value per layer, fixed before threshold tuning: fast in L1, default outside. Values not chosen yet
- Choose it with the Mæhre–Pari timing model (IT BES workshop, slides 20–21): signal scan per candidate value, then cluster count and resolution in CMSSW
- **Always follow a change with a gain scan**



Mæhre and Pari, IT BES workshop, slide 14

Handle: masking

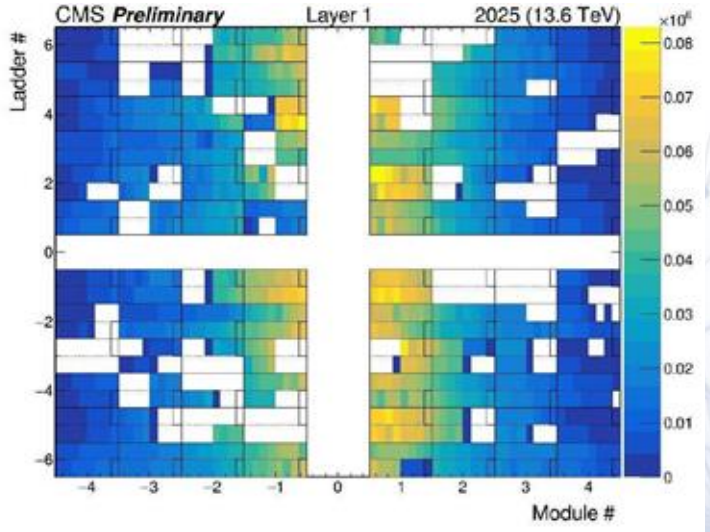
Mask pixels	Noise rate	Hit efficiency	Out-of-time fraction	Single-hit resolution	Collected charge (ToT)
Effect	++	−	–	?	–
Why	Noisy pixels removed	Masked fraction lost	No direct effect	Holes in clusters	No direct effect

+ better · − worse · count = strength · – no effect · ? to check · ↻ redo calibration. Magnitudes are a draft.

- Noisy pixel → fake hits, eats bandwidth. Masked pixel → no hits there
- Three levels: pixel, chip, link/module
- Phase-1 L1: high data rate → link errors → channel auto-masked during the fill. Higher threshold → fewer hits → fewer masked channels

Trigger → action

- Noise scan: hits without injection > 10^{-4} → mask pixel
- PixelAlive: large injection seen < 90% → mask pixel
- Chip stuck → soft reset → hard reset → mask chip
- **Open:** masked budget per layer? Re-check masks at every re-tune



Roguljic (Tracker DPG), IT BES workshop, slide 15

Also to calibrate: VDDA / VDDD trim

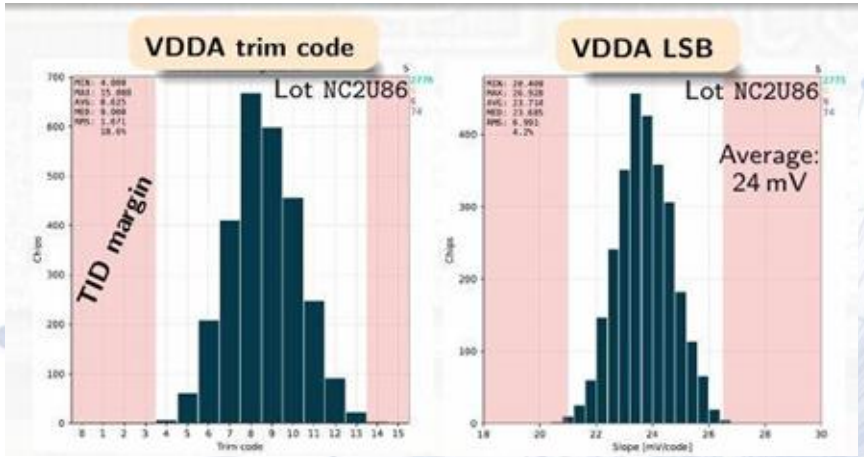
VDD re-trim	Noise rate	Hit efficiency	Out-of-time fraction	Single-hit resolution	Collected charge (ToT)
Effect	↻	?	?	?	↻
Why	Biases move: re-tune threshold	Indirect, via threshold	Indirect, via threshold	Indirect, via threshold	Biases move: redo gain scan

+ better · – worse · count = strength · – no effect · ? to check · ↻ redo calibration. Magnitudes are a draft.

- VDDA → analog front-end and biases. VDDD → digital logic
- Trim (Ph2_ACF voltagetuning): step the code (0–15, ~25 mV), read the rail with the chip ADC, keep the code closest to 1.2 V
- ADC drifts with dose and temperature → recalibration strategies proposed (Gross, Luongo)
- Dose → rails go up (X-rays: +10–15% at 1 Grad; γ: not seen)
- **Above 1.32 V** → the chip ages faster
- **VDDA moves → threshold and gain move**

Trigger → action

- Commissioning → trim each chip, one at a time, cold
- Every ~200 Mrad (L1: ~400 fb^{-1}) → one code down
- Rail close to 1.32 V → re-trim that chip
- VDDA changed → redo threshold and gain



Grippo, IT BES workshop, slide 14

Proposal: calibration programme in time

Calibration	Trigger / cadence	Cost
Full commissioning sequence	Installation; after hardware, cooling or set-point changes	Whole detector, chips in parallel
Threshold + gain + ToT retune	TID steps: 0.5, 1, 5, 10, 50, 100 Mrad, then every 100 Mrad	Largest recalibration cost
VDDA/VDDD re-trim	About every 200 Mrad (to be confirmed)	One chip at a time per serial powering chain
Fast threshold, noise, gain scans	Weekly, on 1/336 of the pixels	Time estimate with DTC ongoing
HV scan	Full once a year; mini every 20–30 fb^{-1}	3–4 h of stable beams (from Ph1)
Timing scan	Once a year, and when the phase monitor drifts	Time estimate with DTC ongoing

Conclusions

- **Radiation moves what we care about**: noise, efficiency, timing, resolution, charge
- For each handle we showed what it moves and how: threshold, delay, HV, Krummenacher, masking; plus the VDDA/VDDD trim
- We proposed a **calibration programme in time**: commissioning → dose steps → monitoring triggers → hand-over to offline
- Threshold: proposal to go from "raise it when forced" to "**as low as the noise budget of each layer allows**"
- Timing with the DTC is speeding up fast: precise time estimates for every calibration in **about two months**

Backup



Online → offline: when to redo CPE

Offline recalibration	Noise rate	Hit efficiency	Out-of-time fraction	Single-hit resolution	Collected charge (ToT)
Effect	–	–	–	++	–
Why	Offline only	Offline only	Offline only	Removes position bias	Offline only

+ better · – worse · count = strength · – no effect · ? to check · ↻ redo calibration. Magnitudes are a draft.

- CPE (Cluster Parameter Estimator) and alignment run offline

Online change → offline action

- HV raised → redo CPE (Lorentz angle, templates)
- Threshold re-tuned → cluster shape changes → check CPE
- Delay changed → cluster size changes → check CPE
- Gain re-scanned → new charge calibration → check CPE
- Pixels/modules masked → update bad-component list

Proposal: observables trigger recalibration

Observable	Source	Action
Min-bias cluster count moves off its minimum	BRIL for TEPX (1 Hz); barrel to be defined	Timing scan
On-track cluster size drops below plateau	DQM	HV scan
Hit residuals drift	DQM, Prompt Calibration Loop	CPE recalibration, alignment
ToT spectrum shifts	BRIL, DQM	Gain scan, HV check
Integrated TID crosses a step	RADSENS, bias currents, simulation	Threshold retune, VDD re-trim
Service-counter rate spikes	Chip auto-read	Soft reset, then hard reset, then mask

The mapping lives in a versioned configuration, not in operator judgment.

Backup: sensor fluence over HL-LHC

	Fluence (1 MeV n_eq/cm^2)
TBPX L1, 3D	3.4×10^{16} by 4000 fb^{-1} → replaced at ~1.8×10^{16}
Planar: TBPX L2+, TFPX	L2: 9.4×10^{15} by 4000 fb^{-1} · TFPX R1: up to 2.3×10^{16} by 4000 fb^{-1} (disks exchangeable)
Consequence	L1 replaced once; TFPX R1 maybe

What we need from the DPG

- Tolerances per layer and ring from simulation: timing shift, threshold dispersion, masked pixels
- A check that cheap global observables (cluster count, ToT) track on-track residuals and Y-resolution
- An online observable for the barrel (TBPX, TFPX)
- Automated HV and timing scans, and a Phase-2 Prompt Calibration Loop
- Automated CPE calibration, maintained by more than one person
- Code in CMSSW, configuration in the conditions database